KaxaNuk

Sharing knowledge

Portfolio Construction & Strategy Modeling

Session 3

Content

- ➔ The Strategy Challenge in Investment Research
- ➔ Strategy Definition
- ➔ Feature Engineering
- ➔ Strategy Design
- ➔ The Tooling and Data Insights Needed

Where We Are in the Research Process

In **Session 1** we discussed:

The Evolution of Investment Research

Observation → Theory → Models → Systems

In **Session 2** we discussed:

Data Curation and Feature Engineering

Raw Data → Structured Data → Features → Signals

Now we answer the question:

How do signals become portfolios?

The Strategy Challenge in Investment Research

Bottlenecks

The Data Challenge

60–80%

of a researcher's time may be wasted on repetitive data cleaning and structuring tasks.

Reproducibility Crisis

- Undocumented workflows make results impossible to validate or replicate.

Scattered Sources

- Inconsistent provider formats require constant, error-prone normalization.

Manual Errors

- Human intervention introduces silent data failures that are hard to detect and harder to fix.

Lost Velocity

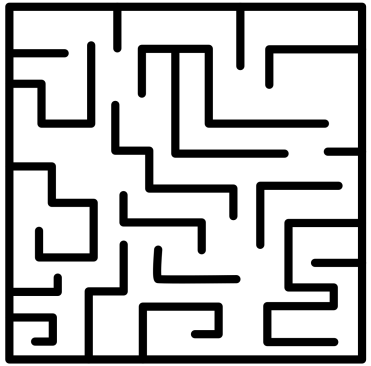
- Every hour spent fixing pipelines is an hour not spent generating alpha.

The Strategy Challenge

Finding a predictive signal is only the beginning.

The real challenge is building a complete strategy system.

Researchers must answer several structural questions:



- What securities can we invest in?
- How do we construct portfolios?
- How do we simulate historical performance?
- How do we understand what drives results?

Investment research is not just about signals.

It is about designing systems that allocate capital.

Signals identify opportunities.

Portfolio construction translates them into capital allocation.

From Signals to Historical Portfolios

Once signals are defined, the next step is constructing historical portfolios.

Historical portfolios allow researchers to simulate:

- Security **selection**
- Position **sizing**
- Rebalancing **timing**

These elements determine how signals translate into portfolio decisions.

Selection, sizing, and timing **can appear in different orders**, and **strategies may use multiple signals within each step**.

The result is a time series of portfolio weights and returns, which becomes the foundation for strategy analysis.

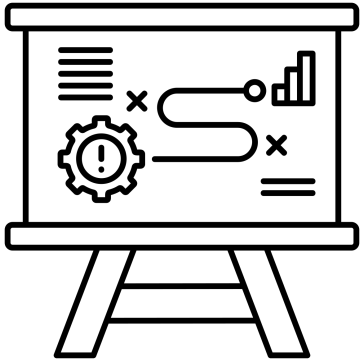
Why Historical Portfolios Matter

A signal alone cannot be evaluated.

We must observe how the signal behaves when applied to real portfolio constraints.

Historical portfolios allow us to answer:

- How concentrated is the strategy?
- How often does it trade?
- How sensitive is it to liquidity?
- How does it evolve across market regimes?



Without portfolio construction, signals remain abstract ideas.

Let's See An Example

From Theory to Practice

Strategy Definition

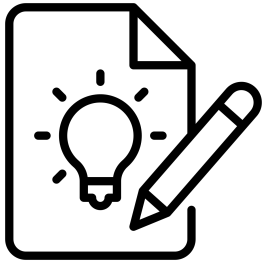
Liquidity-Weighted Trend Strategy

Idea Description

This strategy explores trend persistence in U.S. equities using a simple but robust signal.

The core idea is that stocks experiencing sustained upward price trends tend to continue outperforming due to:

- Gradual information diffusion
- Institutional trading flows
- Behavioral underreaction



Trend persistence is identified through a moving average crossover signal.

Stocks are considered bullish when:

- **SMA(50) > SMA(200)**

Eligible securities are then ranked by 3-month traded value, and the top 35 most liquid stocks are selected.

If fewer than 35 stocks satisfy the trend condition, remaining capital is allocated to:

- **SPY ETF** - SPDR S&P 500 ETF Trust

Background Research

Momentum and trend-following strategies have been extensively documented in financial research.

Key findings from academic literature:

Price Momentum

Returns to Buying Winners and Selling Losers: Implications for Stock Market Efficiency by Narasimhan Jegadeesh and Sheridan Titman (1993)

- Stocks that performed well in the past tend to continue outperforming in the short to medium term.
- One of the most widely replicated anomalies in finance.

Background Research

Trend Following Across Asset Classes

Time Series Momentum by Tobias Moskowitz, Yao Hua Ooi, and Lasse Heje Pedersen (2012)

- Trend-following strategies generate positive returns across equities, commodities, currencies, and bonds.
- Momentum effects persist across global markets.

Behavioral Explanation

Investor Psychology and Security Market Under- and Overreactions by Kent Daniel, David Hirshleifer, and Avanidhar Subrahmanyam (1998)

- Investor underreaction to new information can lead to persistent price trends.

Liquidity and Implementation

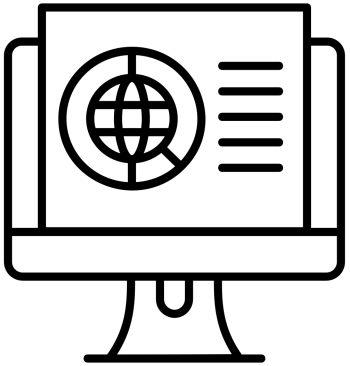
Trading Costs of Asset Pricing Anomalies by Andrea Frazzini, Ronen Israel, and Tobias J. Moskowitz (2014)

- Liquidity constraints and transaction costs are critical when implementing anomaly-based strategies.

Investable Universe and Constraints

Universe:

➔ KN US Equity Benchmark Historical Holdings



Strategy constraints:

- US listed equities
- Minimum trading history required
- Liquidity filter based on traded value
- Long-only portfolio implementation
- Maximum position size limits

These constraints ensure the strategy is realistic and investable.

Benchmark

Benchmark:

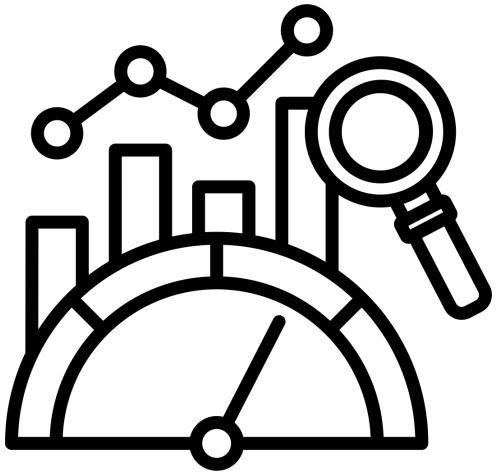
➔ KN US Equity Benchmark

This benchmark provides:

- A transparent investable universe
- Consistent portfolio reference
- Historical benchmark returns

Benchmark comparison allows us to evaluate:

- Excess returns
- Tracking error
- Active performance



Hypothesis

The strategy is motivated by behavioral and structural market inefficiencies.

Behavioral drivers:

- Investors underreact to new information
- Investors chase past performance

Structural drivers:

- Institutional portfolio adjustments occur gradually
- Liquidity concentration reinforces momentum

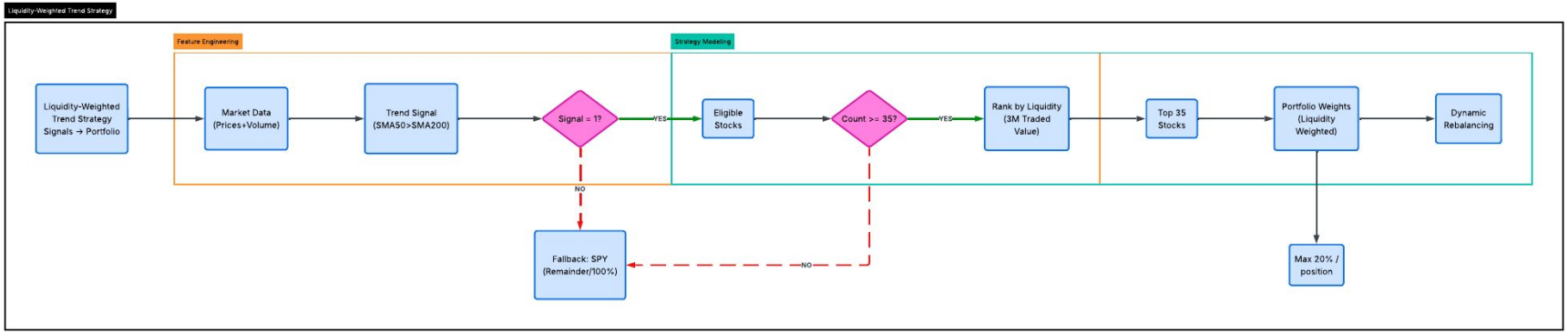
Hypothesis:

Stocks exhibiting sustained positive trends and strong liquidity will outperform the benchmark over intermediate horizons.

Null hypothesis:

The strategy does not outperform the benchmark.

Unified Modeling Language Diagrams



Feature Engineering

Liquidity-Weighted Trend Strategy

Data

The strategy relies primarily on market data.

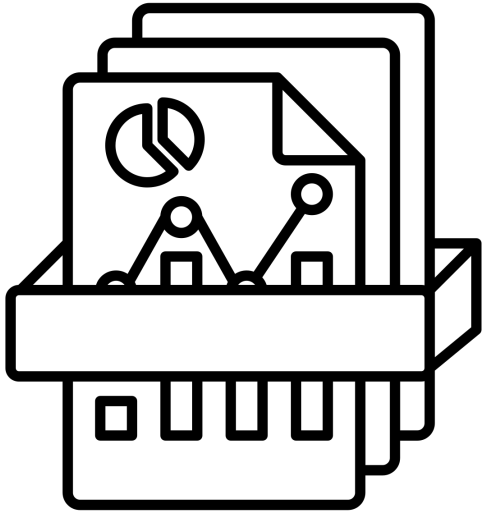
Key variables:

- Adjusted closing prices
- Trading volume
- Traded value

Adjusted prices incorporate:

- Dividends
- Stock splits
- Corporate actions

This ensures signals reflect true total return dynamics.



Data Analysis

Before constructing signals, we examine the statistical properties of the data.

Important observations:

- Equity prices often exhibit trend persistence
- Liquidity is concentrated in a small subset of stocks
- Liquid stocks support scalable systematic strategies

Common visualizations include:

- Price trend charts
- Moving average overlays
- Liquidity rankings

Features

Two core features are constructed from the price data.

Trend Indicators

- c_sma_50d (50-day Simple Moving Average)
- c_sma_200d (200-day Simple Moving Average)

These indicators smooth price movements and capture medium and long-term trends.

Liquidity Feature

- c_daily_traded_value_63d (3-Month Traded Value)

This metric ensures the strategy focuses on highly tradable securities.

Strategy Signals

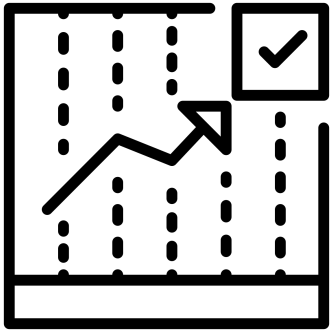
Trend signal definition:

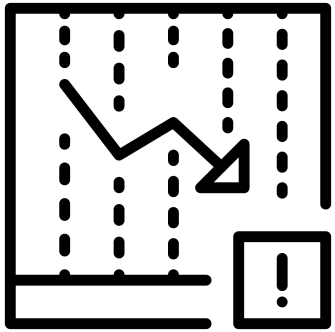
Signal = 1 when

SMA(50) > SMA(200)

Signal = 0 when

SMA(50) ≤ SMA(200)





Implementation in the Data Curator custom calculations **= c_sma_50d_200d_signal**

This rule identifies securities experiencing sustained upward momentum.

Signals are only generated once both moving averages have sufficient historical data.

Strategy Design

Liquidity-Weighted Trend Strategy

Strategy Modeling

Once signals are defined, we must determine how to allocate capital.

Portfolio construction requires answering three key questions.

Selection

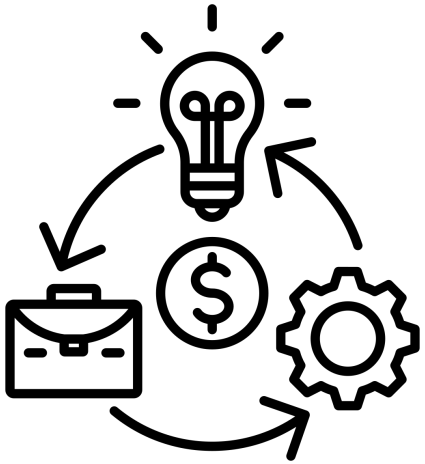
➔ Which stocks are trading above their 50-day and 200-day moving averages and rank among the Top 35 by 3-month traded value?

Sizing

➔ How should capital be allocated across the selected stocks based on their traded value, while respecting the 20% position limit?

Timing

➔ When does the composition of the Top 35 change, requiring the portfolio to rebalance?



Portfolio Construction

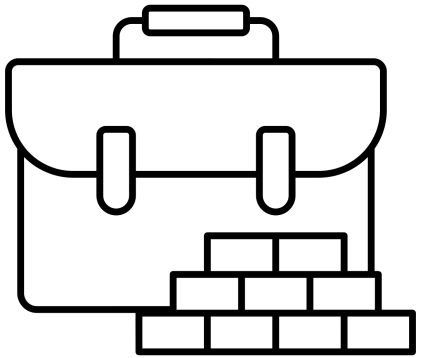
Selection rules:

- Securities with signal = 1
- Ranked by 3-month traded value
- Top 35 securities selected

Sizing rules:

- Position weights proportional to traded value
- Maximum 20% weight per position

These rules define how capital is allocated across securities.



Portfolio Construction

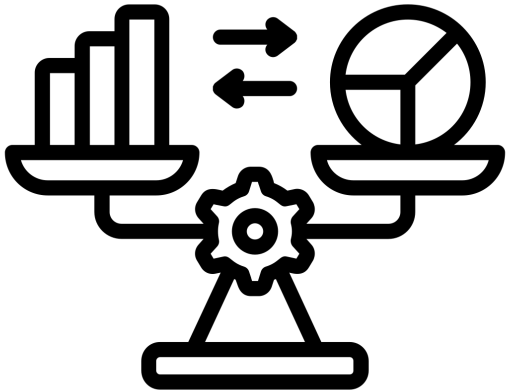
Rebalancing Logic

The strategy uses event-driven rebalancing.

Portfolio changes occur whenever:

- A security enters the Top 35 liquidity ranking
- A security exits the Top 35 liquidity ranking
- The number of eligible securities falls below 35
- A portfolio constituent is delisted or becomes non-tradable

This ensures the portfolio continuously reflects the current signal and investable universe.



Portfolio Construction

Market Regime Handling

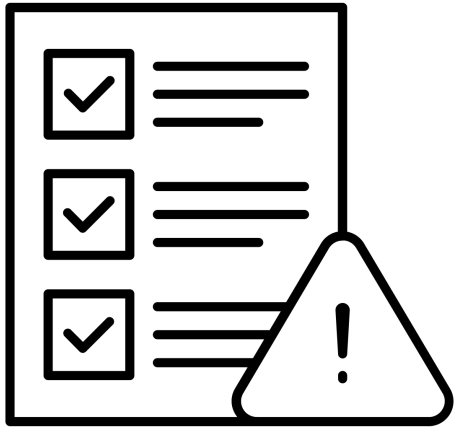
During weak markets, fewer stocks may satisfy the trend condition.

If the number of eligible stocks is less than 35, remaining capital is invested in:

SPY ETF - SPDR S&P 500 ETF Trust

This ensures:

- Capital remains invested
- Market exposure is maintained
- Diversification is preserved



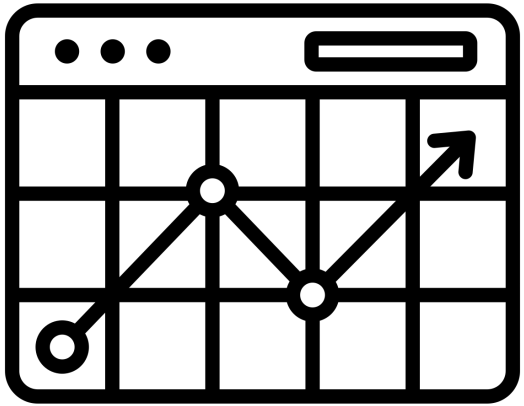
Historical Portfolio Generation

Applying the strategy rules through time produces a time series of historical portfolios.

Each rebalance generates:

- Portfolio constituents
- Position weights
- Daily portfolio returns

These portfolios form the basis for performance evaluation.

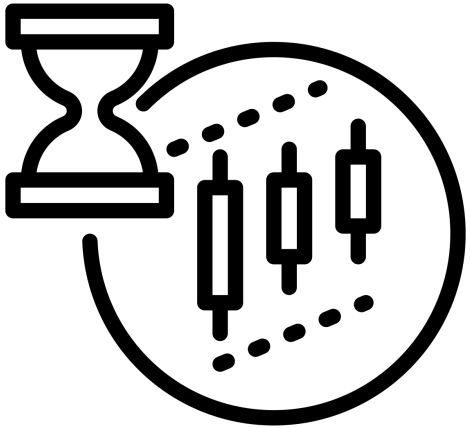


Backtesting

Backtesting evaluates how the strategy would have performed historically.

Key metrics include:

- Cumulative returns
- Volatility
- Drawdowns
- Turnover
- Risk-adjusted returns



However, **performance metrics alone are not sufficient to validate a strategy**.

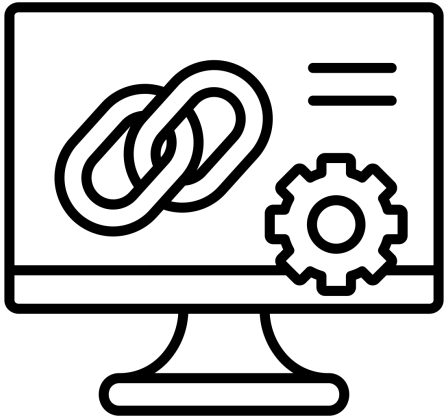
Strategy Robustness

Robust strategies should be evaluated under different scenarios.

Key tests include:

- Out-of-sample performance
- Parameter sensitivity analysis
- Market regime evaluation
- Diversification analysis

These tests help identify strategies that are stable rather than overfit.



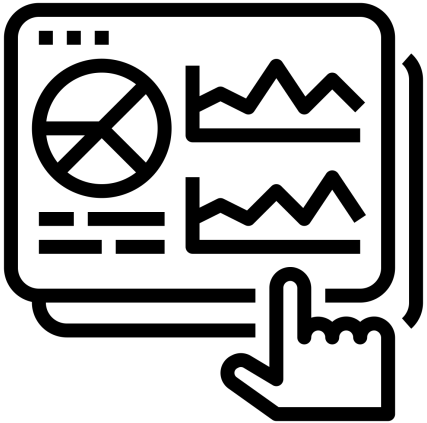
Attribution Analysis

Performance must be decomposed to understand its true drivers.

Key attribution questions:

- How much performance comes from market exposure?
- How much from sector allocation?
- How much from stock selection?
- How much from factor exposures?

Attribution analysis **allows researchers to separate true alpha from unintended risk exposures**.



The Tooling and Data Insights Needed

Software + Analytics

The Portfolio Challenge

Most research stops at:

“Does the signal predict returns?”

But strategy design requires answering three deeper questions during construction:

Selection

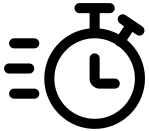
➔ Which securities enter the portfolio?

Sizing

➔ How much capital does each position receive?

Timing

➔ When do we enter, rebalance, or exit?

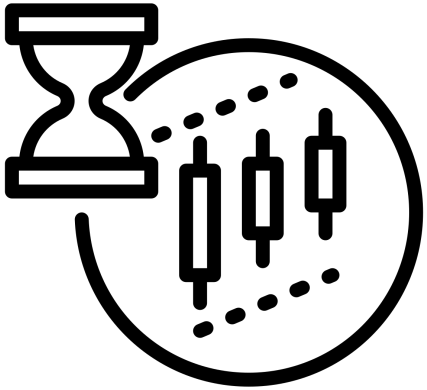


A Backtest Is Only as Good as Its Assumptions

Backtests often look precise.

But they quietly depend on:

- Point-in-time data integrity
- Look-ahead bias controls
- Survivorship bias handling
- Realistic rebalancing logic
- Execution price assumptions
- Transaction cost modeling
- Capital path dependency



Small implementation differences can radically alter results.

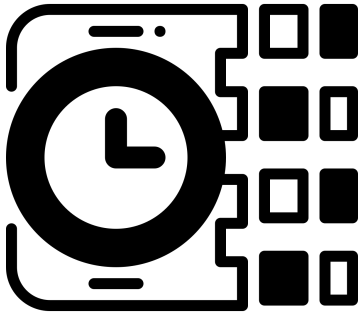
Backtesting should be controlled experimentation, not optimistic simulation.

Returns Must Be Decomposed

Performance Without Decomposition Is Incomplete.

Even when a strategy performs well, key questions remain:

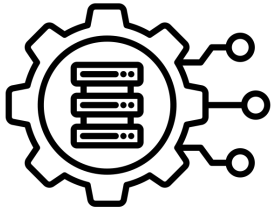
- Is performance coming from market beta?
- From sector tilts?
- From known risk factors?
- From unintended exposures?
- Or from true alpha signals?



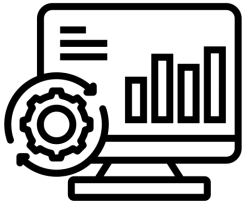
Without structured attribution:

Conviction is misplaced. Risk is misunderstood. Capital allocation becomes fragile.

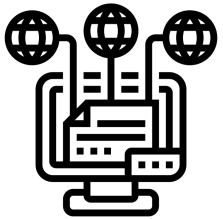
The Tooling To Speed Up Strategy Design



Portfolio Construction



Backtest Engine



Attribution Analysis

https://www.kaxanuk.mx/software

Analytics Layer

Software allows you to test ideas.

Analytics allows you to understand them.

Our Analytics suite provides:

- Transparent benchmark portfolios
- Structured alpha signals
- Purpose-built risk factors

Designed to support professional decision-making.

Benchmark Portfolios

Credible Reference Points for Capital Allocation

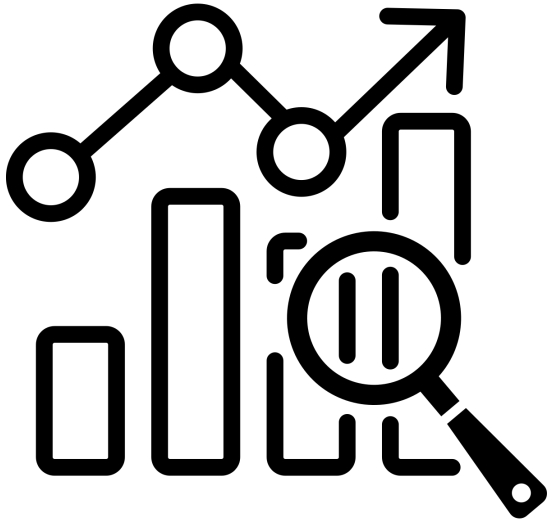
A strategy without a benchmark is directionless.

We build benchmark methodologies that provide:

- Eligible investable universe
- Daily historical weights
- Transparent construction rules
- Full historical return series

Why this matters:

- Compare strategies to realistic baselines
- Avoid “imaginary” or non-investable universes
- Support relative performance evaluation
- Improve governance and reporting



Capital is allocated relative to something. We make that “something” rigorous.

Alpha Signals

Structured, Testable, Deployable

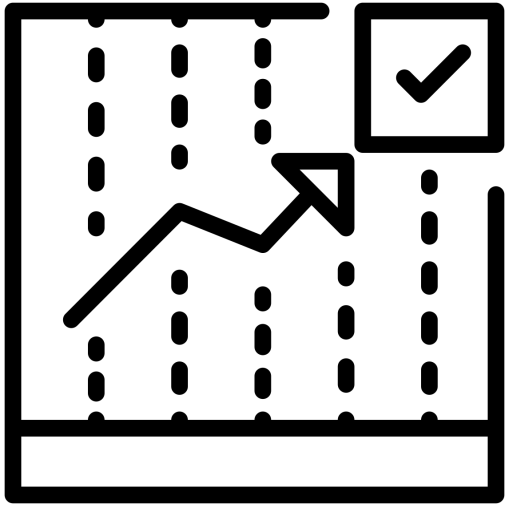
We develop alpha signals by blending:

- Market data
- Fundamental indicators
- Cross-sectional metrics
- Alternative datasets

Each signal is:

- Rules-based
- Fully documented
- Reproducible
- Audit-ready

Designed for validation, not storytelling.



Signals are not opinions. They are engineered hypotheses.

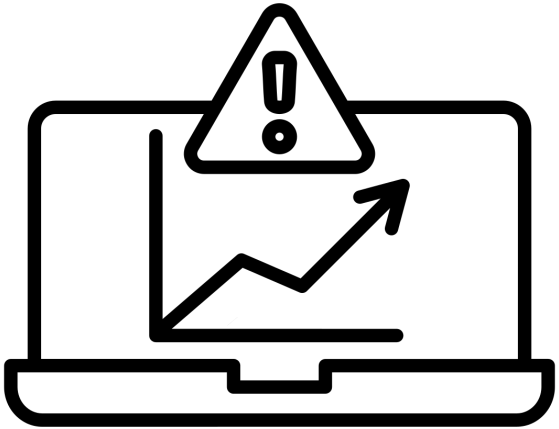
Risk Factors

Understanding Structural Exposures

We design factors specifically for attribution and diagnostics.

Answer clearly:

- Is performance market-driven?
- Is it factor-driven?
- Is it allocation-driven?
- Is it selection-driven?
- Is it residual alpha?



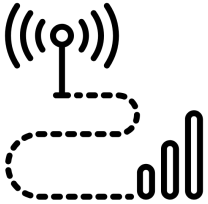
Our factors are:

- Built for cross-sectional decomposition
- Compatible with relative or absolute frameworks
- Structured for portfolio-level analysis

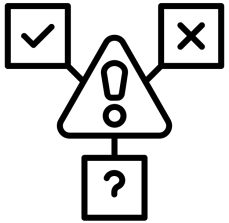
You cannot control what you cannot measure.

Data Insights Into Strategy Performance

Benchmark Portfolios



Alpha Signals



Risk Factors

Let’s code!

Liquidity-Weighted Trend Strategy

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